

The Consistency/Coherence Square

Working note — intuition pump, not a theorem

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Two faces of consistency

Consistency has a syntactic and a semantic face:

- **Syntactic:** No derivation of $\perp$ from the axioms.
- **Semantic:** There exists a model.

The completeness theorem connects them: for first-order logic, these coincide.

Question. Does coherence have two faces? Or is it inherently semantic — the failure mode being non-compact, hence inaccessible to any finitary proof calculus?

The punnet square

Rather than seeking a syntactic face of coherence directly, consider the 2×2 of consistent/inconsistent $\times$ coherent/incoherent:

	Consistent	Inconsistent
Coherent	Classical probability. Local data embeds in one Boolean algebra; CE holds; σ -additive measure exists.	Quantum probability. Incompatible observables (no single Boolean embedding), but Gleason gives a global σ -additive state.
Incoherent	The ultrafilter charge. No contradiction — every first-order test passes — but σ -additivity fails. CE fails.	Kochen–Specker. Non-Boolean (local incompatibility) AND no dispersion-free global state. Both axes fail.

Candidate axes

- **Consistency axis** = distributivity / Boolean embeddability. “Can the local data coexist in a single classical context?”
- **Coherence axis** = state extension / σ -additivity / CE-type admissibility. “Does the local data force a global realization of the target property?”

Programme mapping

The programme’s papers correspond to slices of the square:

- **Paper I:** Top row (Boolean). Distinguishes coherent from incoherent by identifying CE as the admissibility condition.
- **Paper II:** Left column (state exists). Distinguishes consistent from inconsistent — i.e., Boolean (EA/PR/VDR all available) from non-Boolean (VDR blocked by Kochen–Specker).

- **The diagonal** from top-left to bottom-right traces the full filtration: $VDR \rightarrow PR \rightarrow EA \rightarrow \text{nothing}$.

The key tension

The identification of “inconsistency” with “non-Boolean / contextual” is suggestive but not rigorous. Classically, inconsistency means a contradiction is derivable. Here it means something weaker: non-embeddability into a single Boolean context. Whether this is really *inconsistency* or merely *incompatibility* is exactly the question.

Possible resolution: in quantum logic, the failure of distributivity means that the conjunction $a \wedge (b \vee c)$ cannot be decomposed as $(a \wedge b) \vee (a \wedge c)$. This is not a contradiction in the proof-theoretic sense (no $\perp$ is derivable), but it is a failure of the structural rule that makes classical reasoning valid. Perhaps “inconsistency” in the square means: *failure of a structural rule required for classical deduction*, rather than derivability of $\perp$.

Questions for reading

When reading Caramello, Vickers, Kalmbach, Hodges, Pták–Pulmannová:

1. Does the author’s formalism give a definition of coherence?
2. Does the author distinguish consistency from coherence?
3. Where would the author place CE in their framework?
4. Does the non-Boolean case (quantum) fit naturally, or require a separate treatment?
5. Is there a formal sense in which “incompatibility” (non-Boolean) is a species of “inconsistency,” or are they genuinely independent?

Status

Intuition pump. Do not promote to stable files until a definition crystallizes. Hold lightly while reading.